

MARY SEMERDJIAN

Los Angeles, CA | Email | Portfolio | LinkedIn

EDUCATION

Master of Science in Computer Science

California State University Los Angeles, Los Angeles, CA | GPA: 3.9/4.0 | May 2024

- Master's Thesis: "Applying Machine Learning Tools for Real-World DDoS Detection" (Advisor: Dr. Ye)
- Relevant Coursework: Computer and Network Security, Topics in Data Science, Computer Graphics, Computer Networks Configuration and Management, Advanced Topics in Artificial Intelligence, Web Programming, Operating System, Topics in Functional Programming, Advanced Software Engineering, Architecture, Current Trends in Web Design and Development

Bachelor of Science in Computer Science

California State University Los Angeles, Los Angeles, CA | GPA: 3.2/4.0 | May 2022

- Capstone Project: "Pelvic Image Analysis and Geometry Reconstruction using Artificial Intelligence" (Advisor: Dr. Forouzes, Sponsored: Dr. Mathias Brieu (Biomedical))
- Relevant Coursework: Topics in Human-Centered Computing, Cryptography, Advanced Machine Learning and Deep Learning, Concurrent and Distributed Programming, Principles of Database Systems, Theory of Probability
- Dean's List

Associate of Arts in Computer Science, Computer Information Systems, Liberal Arts - Natural Sciences and Mathematics, Arts and Humanities, Associate of Science in Physics for Transfer

Los Angeles City College, Los Angeles, CA | GPA: 3.4/4.0 | June 2019

- Relevant Coursework: Introduction to Computer Science, Programing Logic and Design, Programming in Java and C++, Introduction to Data Structures, Calculus with Analytic Geometry I-II-III, Linear Algebra, Discrete Mathematics, and Physics for Engineers and Scientists I-II-III, Intro to Photoshop, Biology, General Chemistry, Psychology I.
- Dean's List

PROFESSIONAL EXPERIENCE

Graduate Assistant – Computer and Network Security

California State University Los Angeles | Los Angeles, CA | Aug 2023 – Dec 2023

- Supported a graduate-level course, Computer and Network Security course, with 60+ graduated students
- Providing assistance with course material, constructive feedback, focusing on networking topics - network security tool
- Collaborating with the supervisor with additional support and contributing to academic schooling through laboratory work, special projects, and various educational activities
- Grading their assignments, improving the student's understanding of network security and complex cybersecurity concepts

Master Thesis

California State University Los Angeles | Los Angeles, CA | Aug 2023 – May 2024

- Authored a Master's Thesis under the supervision of Dr. Ye, focusing on research related to Distributed Denial of Service
- Conducted an analysis of detection techniques and mitigation strategies for Distributed Denial of Service attacks by leveraging machine learning, statistical algorithms, probability theory, and neural networks to analyze and train datasets effectively
- Acquired experience and developed expertise in machine learning, machine learning frameworks, cybersecurity, and networking
- Utilized both small and large datasets, as well as employing GPUs for testing and training the data
- Prevent the data from overfitting the model

Senior Design Capstone Project

California State University Los Angeles | Los Angeles, CA | Aug 2021 – May 2022

- Led team of 5 undergraduate students in learning the basics of 3-D modeling, using 3D Slicer, 3-D visualization software, and experimenting with Nvidia AIAA
- Collaborating with the documentation lead in the editing and finalizing the finalized drafts in both the software design documents and software requirement documentation
- Focused on creating our own AI model and training it using our data – MRI images of pelvic organs and, lastly, training our own AI model using Clara Train SDK

PROJECTS

Honeypot Server

- Configured a Honeypot server to monitor malicious activity in Kali Linux and Ubuntu
- Attaining insight into potential attacks as well as analyzing network vulnerabilities

Histopathologic Cancer Detection

- Tested and trained different Machine Learning algorithms
- Develop an accurate model for classifying images containing either metastatic cancer cells or non-metastatic cancer cells

Data Science for Business Analytics: Intro to Google Analytics, Adobe Analytics, Microsoft Power BI

- Involved in collecting and analyzing data from multiple sources, applying statistical techniques,
- Utilizing data visualization tools and dashboards to communicate insights and inform strategy
- Leveraged data and communicate findings that drive better decision-making in large - small organizations

MariaDB

- Explored the integration of MariaDB into a Node.js application, exemplified through a book search system example
- Illustrated the practical use of MariaDB, enfolded the additional functionalities, including connecting to the MariaDB, querying, inserting, updating, and deleting data from MariaDB

Artificial Intelligence – Generative Agent

- Reproduce generative agents, a form of Artificial Intelligence (AI) that can simulate human-like behavior based on existing data
 - The approach originates from the research detailed in the "Generative Agents: Interactive Simulacra of Human Behavior" paper

TECHNICAL SKILLS

- **Programming Languages:** C++, C, Java, JavaScript, Python, Haskell, Kotlin, BASIC, Swift
- **Tools & Technologies:** Git, Docker, AWS, Linux, Kali Linux, Ubuntu, Windows, Mac, MySQL, MariaDB, MongoDB, phpMyAdmin, Jupyter Notebook, 3D Slicer, Cisco, Nmap, Metasploit, Snort, Wireshark, Nessus
- **Frameworks/Libraries:** TensorFlow, Node.js, Scikit-learn, Pandas, Django, Bootstrap
- **Soft Skills:** Team Leadership, Problem Solving, Technical Writing, Agile Methodology, Creativity, Critical Thinking